

Module 18 — Practice Tasks

Task 1: `calculateBill(amount, taxRate = 0.05)` — Default Parameters

Return the final bill after tax is applied. Default tax rate is 5% when not provided.

Input:

`calculateBill(1000)`

Output:

1050

Input:

`calculateBill(1000, 0.10)`

Output:

1100

Task 2: `buildProfileCard(user)` — Template Literals

Given `{ name, title, company }`, return a multiline string using template literal interpolation only.

Input:

```
{ name: "Arif", title: "Developer", company: "Programming Hero" }
```

Output:

```
"Arif\nDeveloper at Programming Hero"
```

Task 3: `getLowestPrice(prices)` – Spread Operator

Given an array of numbers, return the lowest price using `Math.min` combined with spread (no manual loop).

Input:

```
[340, 120, 560, 90]
```

Output:

```
90
```

Task 4: `swapValues(a, b)` – Destructuring

Return `[b, a]` using array destructuring assignment, no temporary variable.

Input:

```
swapValues(5, 10)
```

Output:

```
[10, 5]
```

Task 5: `getEmployeeDept(employee)` — Nested Destructuring

Given `{ name, job: { title, department } }`, return `{ title, department }` using nested destructuring.

Input:

```
{ name: "Nadia", job: { title: "PM", department: "Product" } }
```

Output:

```
{ title: "PM", department: "Product" }
```

Task 6: `promoteEmployee(employee, newTitle)` — Spread + Immutable Update

Return a NEW employee object with `title` updated to `newTitle`. Original object must remain unchanged.

Input:

```
promoteEmployee({ name: "Rafi", title: "Junior Dev" }, "Senior Dev")
```

Output:

```
{ name: "Rafi", title: "Senior Dev" }
```

(original object still has title: "Junior Dev")

Task 7: `sumAllPrices(prices)` — for...of

Given an array of numbers, return their sum using `for...of`.

Input: [100, 250, 75]

Output: 425

Task 8: `getManagerName(employee)` — Optional Chaining

Given an employee object that may or may not have a `manager` property, return the manager's name safely, or `undefined` if missing.

Input: { name: "Mim", manager: { name: "Tanvir" } }

Output: "Tanvir"

Input: { name: "Rafi" }

Output: undefined

Task 9: `splitFirstRest(arr)` — Rest Syntax

Return { `first`, `rest` } where `first` is the first element and `rest` is an array of all remaining elements.

Input: [10, 20, 30, 40]

Output: { first: 10, rest: [20, 30, 40] }

Task 10: `cloneAndUpdate(arr, index, newValue)` – Spread
(Immutable Array

Update) Return a new array with the value at `index` replaced by `newValue`, without mutating the original array.

Input: `cloneAndUpdate([5, 10, 15], 1, 99)`

Output: `[5, 99, 15]`

(original array still `[5, 10, 15]`)